

GFT HACKATHON 2026
INTER-DEPARTMENT GROUP FUNCTIONS TECHNOLOGY

Ask your data. Governed.

A standalone ask-your-data service that turns plain-English questions into validated, read-only queries and returns traceable insights.

Team Query Compliance



Governed self-service insights

02 • PROBLEM

Every data question becomes a ticket

1

Translate

Business questions are rewritten as data requests for someone else.

2

Wait

Specialists become the bottleneck for every metric.

3

Debate

Teams argue over inconsistent numbers with no shared source.

WHY DOCUMENT RAG FALLS SHORT

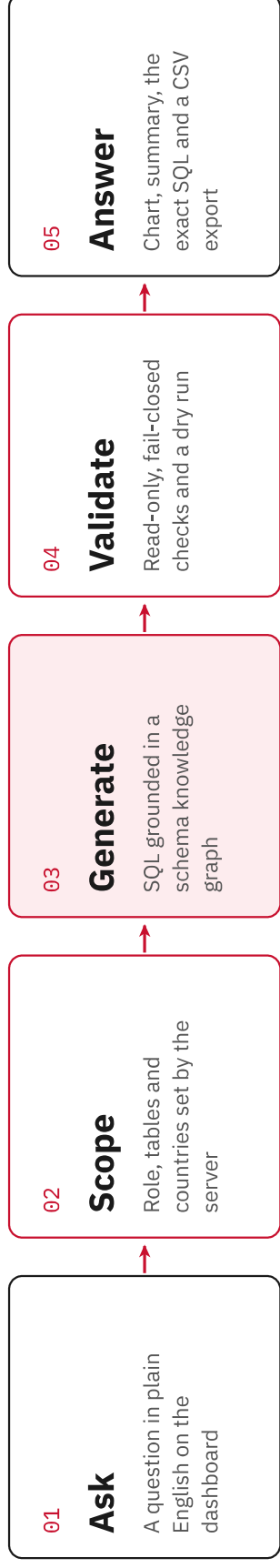
It can **summarize content**, but it can't reliably:

- calculate live metrics
- show where a figure came from
- enforce who can see what

...all while regulated data demands strict control over access.

03 • SOLUTION

Plain English in, traceable numbers out

**Faster decisions**

Answers in one step, not a request queue.

Less specialist dependency

Business users self-serve safely.

Validated, traceable metrics

Every figure carries its query and scope.

04 • BUSINESS VALUE

From a ticket queue to one self-serve question



5 → 1 hand-offs

Minutes not a queue

Self-serve routine KPIs

1 source of truth

TIME BACK FOR EVERY STAKEHOLDER

Business users
Ask directly; chart, summary and CSV in one session.

Answers on demand

Business analysts
Routine requests self-served; focus on deeper analysis.

Fewer routine tickets

Compliance & risk
Every answer carries SQL, job ID and access scope.

Audit-ready by default

Management
One governed source; consistent, current figures.

Faster decisions

What a business user sees

ACCOUNT & PERMISSIONS	Your profile		Sign out	
	Username	Role	Access level	
	Investigation	Investigation	Read-only analytics	
	Permitted tables			
COUNTRIES	4 tables		Transactions	
	Explainability	Country	Code	Entity
	Parity	Hong Kong	HK	M&E
	RedCard count	France	FR	H&BC
	RiskScores	United Kingdom	GB	H&BC
		Ireland	IE	H&BC
		India	IN	H&BC
		Poland	PL	H&BC
		Saudi	TW	H&BC
		Thailand		

- 1 Sign in by role:** monitoring, investigation or admin
- 2 Profile** lists the exact tables and countries allowed
- 3 Ask** and get a chart, summary and one-click CSV export

06 • DATASET & SCHEMA

Built on Google's AML AI data model

1,000

retail customers

50,000

transactions

12

BigQuery tables

7

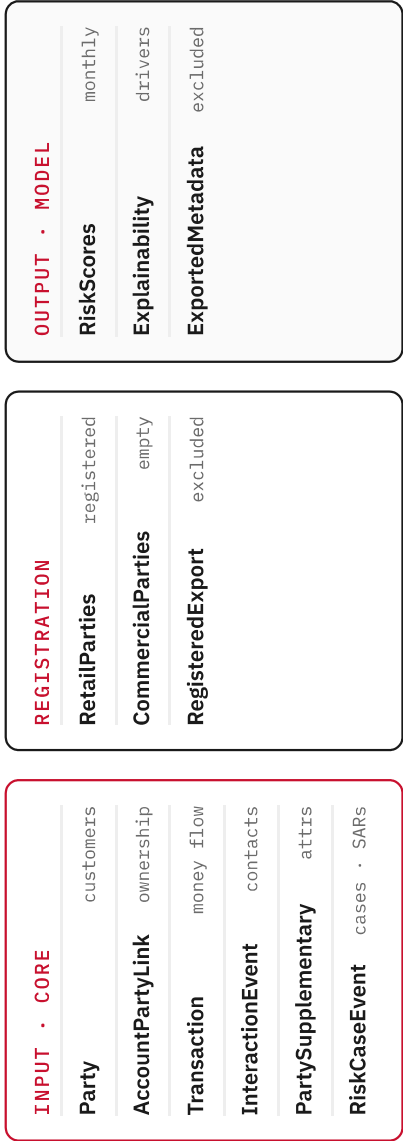
entity-country scopes

8 mo

Jan – Aug 2026

120

investigations • 45 SARs



Nested STRUCT / ARRAY fields, enums and observed values are documented per field, so the graph and validator know exactly what exists.

Google
Big Query

Served via API

by the DRA team

aml_demo

asia-south1

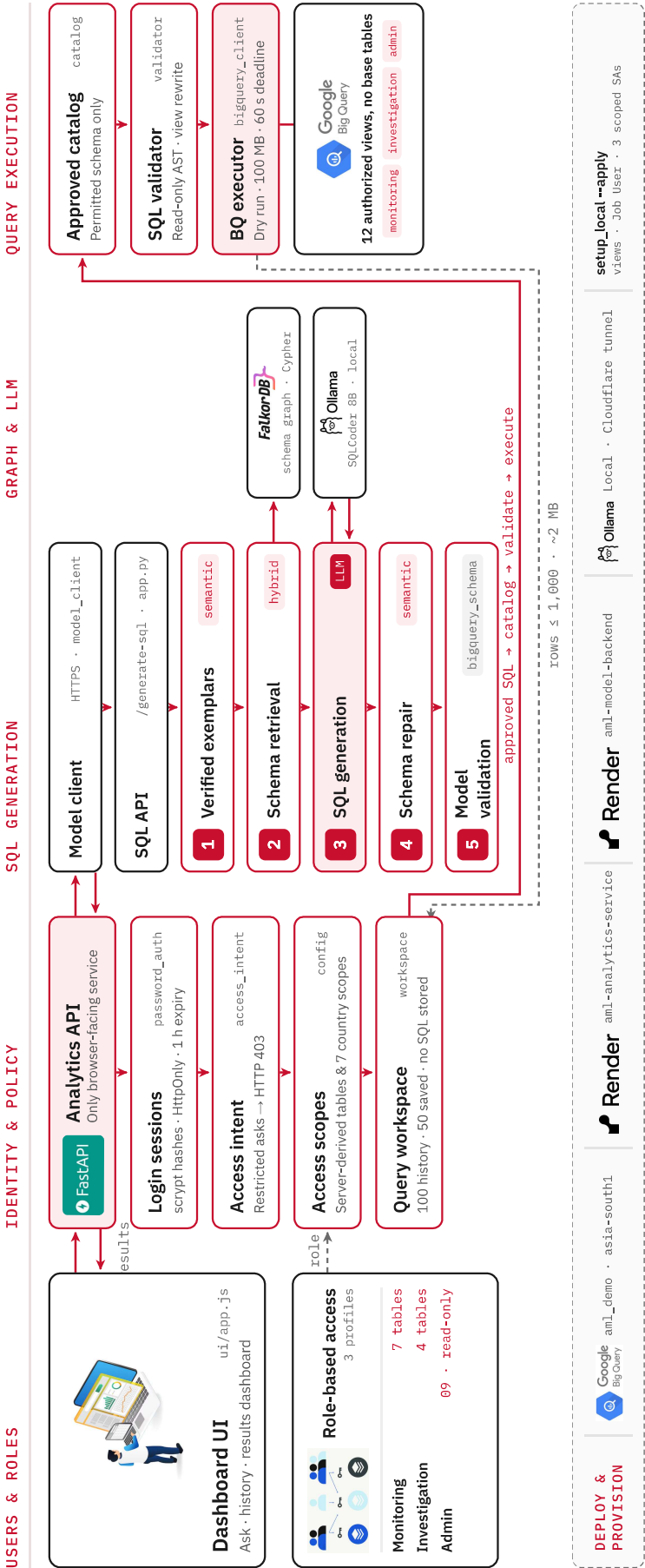
Synthetic data

fictional people and outputs

07 • SYSTEM ARCHITECTURE

Governed Text-to-SQL for AML analytics

Role-scoped questions become validated, read-only BigQuery results, generated by a local LLM grounded in a schema graph.



Server-side RBAC

Role and scope come from the session, never the request.

Authorized views only

Each role's service account reads only its own views.

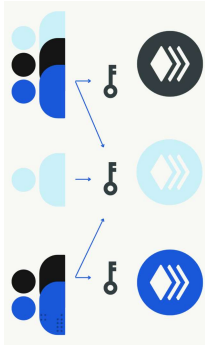
Fail-closed execution

AST check, dry run and scan budget before any query runs.

Grounded, local LLM

FalkorDB schema slice feeds SQLCoder on Ollama.

The server decides who sees what, not the prompt



ROLE	AUTHORIZED VIEWS
Monitoring	Party, Transaction, AccountPartyLink, InteractionEvent + 3 registration tables
Investigation	Party, RiskCaseEvent, RiskScores, Explainability
Admin	Union of both (10 tables), still read-only

All roles: 7 entity-country scopes. Export tables are excluded for everyone.

SIX LAYERS, ALL MUST PASS

- 1

Session login
scrypt hashes • HttpOnly, SameSite=Strict • 1 h • 10 sign-ins/min
- 2

Access intent
Clear restricted asks return HTTP 403 before generation
- 3

Permitted schema only
The model never sees tables outside the role
- 4

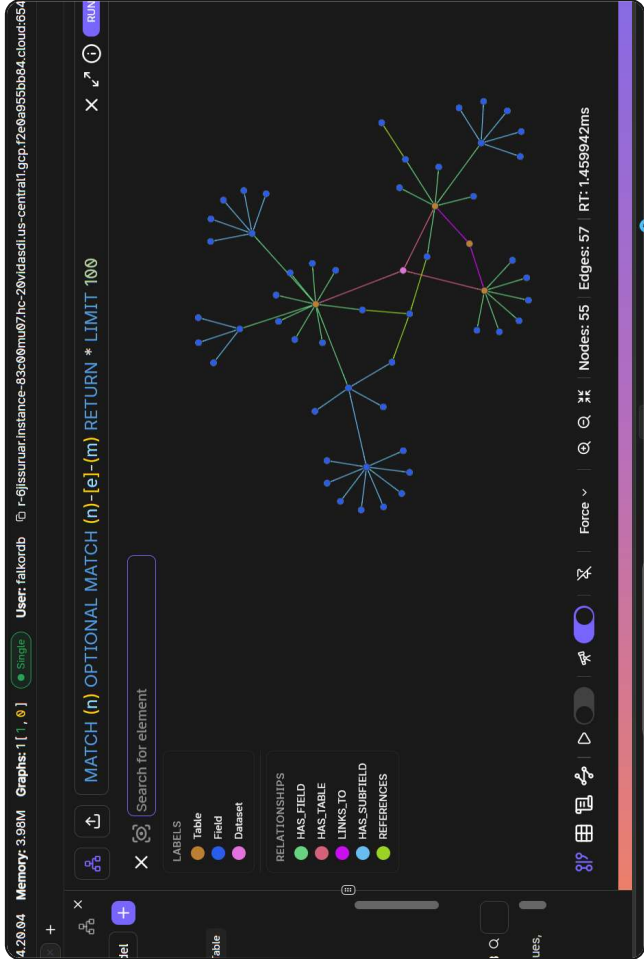
Independent SQL validator
One read-only statement • allowlists • rewrite to views
- 5

BigQuery dry run & budgets
100 MB scan • 60 s deadline • 1,000 rows / ~2 MB
- 6

Per-role service account
Data Viewer on its views only • no base-table access

09 • GROUNDING & ACCURACY

Grounded in a schema graph, not a static prompt



FalkorDB Table • Field • Dataset • Metric nodes, joined by key edges

Query Compliance • GFT Hackathon 2026

Hybrid retrieval

Exact plus semantic matching: "customers" finds Party, "SAR" finds RiskCaseEvent. Joinable tables come along.

Verified exemplars first

Recurring KPI questions reuse approved SQL instead of regenerating.

Local, deterministic LLM

SQLCoder 8B on Ollama at temperature 0. No fine-tuning needed.

Two independent checks

Repair against retrieved schema, then validate against the full model.

Every number shows its working

▼ Analysis details

Synthetic demo data; risk outputs are simulated.
KPI definitions require domain-owner approval before real banking use.
AI-generated SQL; passed schema, access and execution checks. These checks do not prove it matches your intended business meaning.

Request	b9102dbb-48a8-4a8a-b1bc-c89b29f0f836
Job	aml_b9102dbb48a84a8ab1bcc89b29f0f836
Mode	bigquery
Scope	HASE_HK_HSBC_FR_HSBC_GB_HSBC_IE_HSBC_IN_HSBC_PL_HSBC_TW
Bytes scanned	303696
Schema	50b86516f0c7b31f5a37f0d2df265cd7ec0af9b4b8d926e465e92b3243be5ee6

SELECT _0.type AS type, COUNT(*) AS transaction_count FROM `gen-lang-client-0810987953.aml\_analytics\_demo.Transaction` AS _0 GROUP BY type ORDER BY transaction_count DESC

Executed SQL

Bytes scanned

BigQuery job ID

Schema version

Access scope

Live on BigQuery: 1,000 customers • 50,000 transactions • 12 tables • 7 countries

synthetic data

Mapped to the brief

MVP REQUIREMENT	WHAT WE BUILT	STATUS
Natural-language querying	Plain-English questions over approved BigQuery views	Delivered
Validated query generation	Graph-grounded SQLCoder, repair and independent validation	Delivered
Read-only guardrails	Single read-only statement, dry run, scan and time budgets	Delivered
Source / query visibility	SQL, job ID, bytes, schema version and scope per answer	Delivered
Authorization aligned to access	Per-role views and service accounts across 7 country scopes	Delivered
Basic exports (CSV)	One-click CSV of the authorized result set	Delivered
In-house visualization	Charts and graphs built from each validated result	Delivered
BI links (Qlik Sense, Looker)	Not supported yet; approved outputs to publish next	Next
NoSQL sources	BigQuery first; same gate extends to new sources	Roadmap

12 • DEPLOYMENT & ROADMAP

Live today, governed next

NOW • LIVE POC**Render** 2 services

Analytics service: login, dashboard, CSV export

Model service: internal SQL API

SQLCoder on local Ollama via tunnel

BigQuery via 3 scoped service accounts

NEXT • GOVERNED

Cloud Run with Secret Manager

VPC Service Controls perimeter

Redacted, retained audit log

Row-level policies per user

Excel export and BI connectors

THEN • AI FEATURES

Ambiguity and anomaly detection

Recurring insight packs

Publish approved outputs to BI

NoSQL sources behind the same gate

Thank you

Team Query Compliance

Live demo
aml-analytics-service.onrender.com